

BeamLock

Physics-Informed Virtual Coarse-to-Fine PAT
for Mobile Free-Space Optical Communication

Technical Deep Dive, Results, and Oral Briefing Notes

Team Cadets

Smart India Hackathon 2026 · Problem ID SIH26169

Theme: Smart Automation · Category: Software

Repository: <https://github.com/MudithDShetty/SIH>

September 11, 2026

Abstract

BeamLock (repository name: *FSOC Virtual Tracker*) is a real-time software sandbox for Pointing, Acquisition, and Tracking (PAT) of mobile free-space optical (FSOC) links. It closes a coarse AZ/EL camera loop under a statistical atmospheric channel ($C_n^2 \rightarrow$ Fried r_0 , beam wander, scintillation), Gaussian-beam link SNR, Classical or AI+fusion beacon detection, constant-velocity Kalman tracking, raster re-acquisition when lock is lost, and a readiness-gated mock fine stage (4-quadrant detector + FSM). This document explains the full system architecture, governing equations as implemented in code, control logic, evaluation methodology, verified numerical results, honest limitations, and a structured 20-minute briefing script. It is intended for SIH evaluators and technical reviewers.

Contents

1	Problem Statement and Scope	2
1.1	SIH problem	2
1.2	Operational need	2
1.3	What BeamLock is and is not	2
2	System Architecture	2
2.1	End-to-end data flow	2
2.2	Primary modules	3
2.3	Scenario presets	3
3	Angular Geometry and Camera Plant	3
3.1	Instantaneous field of view (IFOV)	3
3.2	AZ/EL plant	4
4	Atmospheric Channel Physics	4
4.1	C_n^2 mapping from UI strength	4
4.2	Default optical / geometric assumptions	4
4.3	Fried parameter r_0	4
4.4	Rytov variance	4
4.5	Beam wander / tip-tilt RMS	4
4.6	Scintillation irradiance factor I	5
4.7	Appearance coupling	5

5	Optical Link Budget and Link Readiness	5
5.1	Received power model	5
5.2	Geometric collection	5
5.3	Gaussian pointing loss	5
5.4	SNR and fade margin	5
5.5	Link Readiness score	6
6	Detection, Tracking, and Re-acquisition	6
6.1	Classical detector	6
6.2	AI detector	6
6.3	AI+fusion (live “AI” path)	6
6.4	Kalman tracker	6
6.5	Tracking state machine	7
6.6	Re-acquisition	7
7	Fine Pointing Stage	7
7.1	Handoff criteria	7
7.2	4-QD mock	7
8	Evaluation Methodology	7
8.1	Matched-seed Classical vs AI	7
8.2	Acquisition Monte Carlo	8
8.3	Clutter Pd/Pfa	8
8.4	Ablation / packaging	8
9	Verified Results	8
9.1	Classical vs AI scenario matrix	8
9.2	Acquisition Monte Carlo (file-backed)	8
9.3	Classical Stress clutter	8
9.4	Numbers deliberately excluded	8
10	Standards Alignment and References	9
11	Limitations and Honest Claims	9
12	20-Minute Briefing Script (Detailed)	9
12.1	Timing map	9
12.2	Spoken script (expanded)	10
12.3	Hard Q&A crib	10
13	Reproducibility Commands	11
14	Conclusion	11

1 Problem Statement and Scope

1.1 SIH problem

ID: SIH26169.

Title: Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals.

Theme: Smart Automation · **Category:** Software · **Team:** Cadets.

1.2 Operational need

FSOC links use narrow laser beams. Before a fine pointing stage (fast steering mirror / FSM and quadrant detector / 4-QD) can close the communications beam, a *coarse* stage must:

1. observe a remote beacon with a wide-field camera;
2. detect the beacon under turbulence, vibration, and sensor noise;
3. estimate its motion;
4. continuously slew the camera/gimbal so the beacon remains in the field of view (FOV).

Hardware PAT benches are costly and slow for algorithm iteration. BeamLock provides a laptop-scale virtual camera and closed-loop control environment with Classical vs. AI comparison and reproducible logs.

1.3 What BeamLock is and is not

Is	Is not
Real-time coarse PAT simulator (~ 60 fps)	Flight-qualified ISRO/SDA software
Physics-informed statistical channel + SNR handoff score	Certified BER / outage calculator
Classical and AI+fusion detectors in-loop	Split-step / phase-screen wave-optics propagator
Raster re-acquisition + Fine 4-QD mock	Optical hardware-in-the-loop
CSV/JSON + Streamlit PDF evidence	A full communications modem

2 System Architecture

2.1 End-to-end data flow

1. **UI / presets:** Calm, UAV, Stress; sliders for turbulence, vibration, noise, slew; detector toggle; multi-beacon.
2. **Scene:** one or more moving beacons (linear / circular / random walk).
3. **Disturbances + physics:** C_n^2 channel, platform vibration AR(1), sensor noise; Gaussian link budget.
4. **VirtualCamera:** FOV crop, AZ/EL plant with rate/acceleration limits, true IFOV.
5. **Detection:** Classical *or* AI+fusion (mutually exclusive live path).
6. **Kalman tracker:** state LOCKED / COASTING / LOST; camera driven by estimate, not ground truth.

7. **Link Readiness:** physics-informed score in $[0, 1]$; LOST forces 0.

8. **Control:**

- if LOST $\rightarrow$ raster re-acquisition waypoints + distance-scaled slew;
- else if Fine active $\rightarrow$ mock 4-QD / FSM residual nulling;
- else $\rightarrow$ coarse AZ/EL slew toward Kalman estimate.

9. **Outputs:** μ rad-primary HUD, CSV + run-sheet JSON, Streamlit report/PDF, offline Monte Carlo / Pd-Pfa / ablation scripts.

2.2 Primary modules

Module	Role
scene/	Beacon motion, virtual world render
scene/camera.py	AZ/EL plant, FOV, IFOV
disturbance/	Warp residual, vibration, sensor noise coupling
physics/atmosphere.py	$C_n^2 \rightarrow r_0$, Rytov, wander, scintillation
physics/link.budget.py	P_{rx} , SNR, fade margin, pointing loss
detector/	Classical, YOLO, Hybrid fusion, preprocess
tracker/kalman_tracker.py	Constant-velocity KF + state machine
control/reacquisition.py	Raster (default) search
control/fine.pointing.py	Readiness-gated Fine mock
metrics/	Link readiness, CSV/JSON logging
ui/	Controls and scenario presets
report.py	Streamlit analysis + PDF
main.py	Live closed loop

2.3 Scenario presets

Name	Turb	Vib	Noise	Slew	Trajectory
Calm (F1)	0	0	0.05	90	Circular
UAV (F2)	4	8	0.2	90	Circular
Stress (F3)	7	15	0.35	120	Random walk

Demo keys: T Classical $\leftrightarrow$ AI+fusion, L force detector blind, B 1 $\leftrightarrow$ 2 beacons.

3 Angular Geometry and Camera Plant

3.1 Instantaneous field of view (IFOV)

Default acquisition FOV is scaled to the ESA IZN-1 class ~ 2.5 mrad horizontal FOV:

$$\text{IFOV} = \frac{\text{FOV}_H}{N_x} = \frac{2500 \mu\text{rad}}{400 \text{px}} = 6.25 \mu\text{rad/px}. \quad (1)$$

Pixel error maps to angular error by

$$\theta_{\mu\text{rad}} = e_{\text{px}} \cdot \text{IFOV}. \quad (2)$$

3.2 AZ/EL plant

The virtual camera maintains boresight (x, y) in scene coordinates with rate/acceleration limits (slew rate exposed in UI as px/s and equivalently $\mu\text{rad/s}$ via IFOV). When tracking is healthy, `point_towards` drives toward the Kalman estimate. During LOST search, slew multipliers increase with distance to the current waypoint (capped by `MAX_ACQ_SLEW_MULTIPLIER` in `main.py`).

4 Atmospheric Channel Physics

Implementation: `physics/atmosphere.py`. References: Andrews & Phillips style order-of-magnitude models for horizontal / UAV-class paths. **Not** a split-step phase-screen propagator.

4.1 C_n^2 mapping from UI strength

Slider strength $s \in [0, 10]$ maps log-uniform onto $[C_{n,\min}^2, C_{n,\max}^2]$ with $C_{n,\min}^2 = 10^{-17}$, $C_{n,\max}^2 = 10^{-12} \text{ m}^{-2/3}$:

$$C_n^2(s) = \begin{cases} C_{n,\min}^2, & s \leq 0, \\ 10^{\log_{10} C_{n,\min}^2 + (\log_{10} C_{n,\max}^2 - \log_{10} C_{n,\min}^2)(s/10)}, & s > 0. \end{cases} \quad (3)$$

Interpretation: calm $\sim 10^{-17}$, moderate $\sim 10^{-14}$, strong $\sim 10^{-12}$.

4.2 Default optical / geometric assumptions

Symbol / quantity	Default
Wavelength λ	976 nm (acquisition-beacon class)
Path length L	8000 m (UAV / short terrestrial)
Receive aperture D	0.08 m

4.3 Fried parameter r_0

Plane-wave form used in code:

$$r_0 = (0.423 k^2 C_n^2 L)^{-3/5}, \quad k = \frac{2\pi}{\lambda}. \quad (4)$$

Smaller r_0 implies worse atmospheric coherence / seeing.

4.4 Rytov variance

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}. \quad (5)$$

Used as the driver for scintillation sampling regime.

4.5 Beam wander / tip-tilt RMS

One-axis RMS wander (μrad) blends an angle-of-arrival scale $\sim 0.5\lambda/r_0$ with mild aperture averaging and a C_n^2 -dependent floor so mid-strength turbulence remains visible:

$$\sigma_{\text{AoA}} \approx 0.5 \frac{\lambda}{r_0} \cdot A(D, r_0), \quad (6)$$

$$\sigma_{\text{floor}} \propto \sqrt{C_n^2} \sqrt{L}, \quad (7)$$

$$\sigma_{\text{wander}} = \max(\sigma_{\text{AoA}}, \sigma_{\text{floor}}). \quad (8)$$

Temporal correlation uses an AR(1) update with coefficient $\alpha \approx 0.85^{60\Delta t}$ (Greenwood-ish tens-of-ms memory).

4.6 Scintillation irradiance factor I

Irradiance multiplier with mean ≈ 1 :

- Weak ($\sigma_R^2 < 0.5$): lognormal sampling with $\sigma_I^2 \approx \sigma_R^2$.
- Moderate/strong: gamma-gamma style product of two unit-mean gammas (simplified Andrews α, β mapping), clamped to a practical dynamic range for the 60 fps loop.

Scintillation is temporally smoothed more slowly than tip/tilt (aperture-averaging intuition).

4.7 Appearance coupling

Wander offsets the drawn beacon centroid in scene pixels. Scintillation and pointing loss modulate spot intensity. Turbulence and sensor-noise models further degrade the camera-view image before detection.

5 Optical Link Budget and Link Readiness

5.1 Received power model

Implementation: `physics/link_budget.py`. Directed FSO form (cf. IISc/QOSMIC framing, arXiv:2507.20908):

$$P_{rx} = P_{tx} \eta_{tx} \eta_{rx} G_{\text{geo}} T_{\text{atm}} L_{pe} I_{\text{scint}}. \quad (9)$$

Default parameters (simulation, not a flight terminal datasheet): $P_{tx} = 0.5 \text{ W}$, $\eta_{tx} = \eta_{rx} = 0.7$, $D_{rx} = 0.08 \text{ m}$, divergence half-angle $\theta_{\text{div}} \approx 2.6 \text{ mrad}$, $L = 8 \text{ km}$, static $T_{\text{atm}} = 0.75$, NEP-like noise floor $5 \times 10^{-9} \text{ W}$.

5.2 Geometric collection

$$G_{\text{geo}} = \min\left(1, \frac{\pi(D_{rx}/2)^2}{\pi(\theta_{\text{div}}L)^2}\right). \quad (10)$$

5.3 Gaussian pointing loss

With residual angle θ (rad):

$$L_{pe} = \exp\left(-2\left(\frac{\theta}{\theta_{\text{div}}}\right)^2\right). \quad (11)$$

5.4 SNR and fade margin

$$\text{SNR}_{\text{lin}} = \frac{P_{rx}}{P_{\text{noise}}}, \quad (12)$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}_{\text{lin}}, \quad (13)$$

$$M_{\text{fade}} = \text{SNR}_{\text{dB}} - 10 \text{ dB}. \quad (14)$$

The 10 dB term is a handoff floor used for readiness, **not** a claimed BER threshold.

5.5 Link Readiness score

Implementation: `metrics/link_readiness.py`. Physics path (default in `main.py`):

$$S = \text{clamp}(0.45 s_{\text{SNR}} + 0.25 L_{pe} + 0.15 \min(1, I) + 0.15 s_{\text{lock}}, 0, 1), \quad (15)$$

with hard rule

$$\text{if tracking state} = \text{LOST then } S = 0. \quad (16)$$

s_{SNR} maps roughly 10 dB $\rightarrow$ 0, 25 dB $\rightarrow$ 1. **Purpose:** coarse-to-fine handoff decision support.

Not: certified BER, CCSDS outage, or flight link closure certificate.

6 Detection, Tracking, and Re-acquisition

6.1 Classical detector

Grayscale threshold (default 200) $\rightarrow$ largest contour centroid. Baseline for a bright beacon on dark sky. Adaptive mode available under elevated sensor noise.

6.2 AI detector

YOLOv8n trained on synthetic frames generated from the same disturbance pipeline (`scripts/generate_train_scripts/train.py`). Weights: `weights/beacon_yolov8n.pt`. Returns highest-confidence box centroid.

6.3 AI+fusion (live “AI” path)

HybridDetector:

1. denoise / top-hat enhance;
2. YOLO on raw + enhanced;
3. adaptive classical on enhanced;
4. fuse by consensus (~ 35 px) or best candidate;
5. short temporal hold (8 frames) during gaps.

Raw YOLO alone is fragile in heavy grain; fusion is the production live path when AI is selected.

6.4 Kalman tracker

State $\mathbf{x} = [x, y, v_x, v_y]^\top$ with constant-velocity transition

$$F = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \quad (17)$$

Process noise Q uses the standard discrete white-noise acceleration structure scaled by process-noise intensity. Measurement noise $R = \sigma_z^2 I_2$.

6.5 Tracking state machine

State	Condition
LOCKED	Valid accepted detection this frame
COASTING	1–15 consecutive missed measurements
LOST	> 15 misses (or forced lost on detector switch)

Camera control uses the Kalman estimate when not LOST. On LOST, re-acquisition owns the slow target.

6.6 Re-acquisition

ReacquisitionController:

- **Default mode on activate: raster** (covariance-scaled step spacing).
- Spiral waypoint generator exists in code but is not the default activate path (spiral corner-trapped under high error).
- Search center biased by last lock + last velocity + lost time.
- Plausibility gating rejects false re-locks far from search/boresight while LOST.

7 Fine Pointing Stage

Implementation: `control/fine_pointing.py`.

7.1 Handoff criteria

Parameter	Value
Enter readiness	$S \geq 0.70$
Exit readiness	$S \leq 0.55$ (hysteresis)
Tracking	must be LOCKED to enter
Capture cone enter	residual $\leq 600 \mu\text{rad}$
Capture drop	residual $> 900 \mu\text{rad}$
Stable frames	12 (~ 0.2 s at 60 fps)
Fine slew authority	4000 $\mu\text{rad/s}$ class
Quad noise / FSM latency	mild demo non-idealities

7.2 4-QD mock

Spot location relative to FOV center yields normalized quadrant imbalance (q_x, q_y) and residual angle. The controller nulls residual with high-bandwidth tip/tilt relative to the coarse plant. This is a **gated demo plant**, not a calibrated flight FSM/4-QD identification.

8 Evaluation Methodology

8.1 Matched-seed Classical vs AI

`scripts/run_comparison.py` reseeds per-frame RNG so turbulence/vibration/noise realizations match across detectors. Primary metrics: lock retention %, re-acquisition count, mean/max pixel error, readiness, acquisition time.

8.2 Acquisition Monte Carlo

`scripts/monte_carlo_acquisition.py` vs ESA ESTOL warm-start bound $T_{\text{acq}} < 60\text{s}$. Reports median, mean, p95, pass count.

8.3 Clutter Pd/Pfa

`scripts/evaluate_detection.py` under Stress-like conditions. Cite Classical numbers only when AI/hybrid JSON shows pathological $P_{fa} = 1$.

8.4 Ablation / packaging

`scripts/run_ablation.py`, Streamlit PDF from `report.py`, `scripts/build_exe.py` for standalone packaging. Demo video: `scripts/record_demo_video.py` $\rightarrow$ `logs/demo/beamlock_sih_demo.mp4`.

9 Verified Results

9.1 Classical vs AI scenario matrix

Source: `docs/technical_report.md` (matched seed $\approx 45\text{s}$, seed 42, after hard-curriculum re-train).

Scenario	C err	AI err	C lock	AI lock	Reacq C/AI
Calm	35.5 px	35.5 px	100%	100%	1 / 1
UAV	35.6 px	35.6 px	97.7%	100%	1 / 1
Stress	35.1 px	36.1 px	95.5%	95.5%	2 / 1

Interpretation. Mean pixel error remains $\sim 35\text{ px}$ even in Calm: largely **slew-lag limited**. AI’s documented advantage is lock retention (UAV) and fewer re-acquisitions (Stress), not mean-error domination. Do not claim Stress mean-error superiority.

9.2 Acquisition Monte Carlo (file-backed)

Scenario	Trials	Acquired	ESTOL<60s	Median T_{acq}	p95
UAV	40	40/40	40/40	0.817 s	0.85 s
Stress	40	40/40	40/40	0.967 s	1.0 s

Files: `logs/monte_carlo/acquisition_uav_40trials.json`, `logs/monte_carlo/acquisition_stress_40tri`

9.3 Classical Stress clutter

From `logs/detection/pd_pfa_stress.json`: $P_d \approx 0.667$, $P_{fa} \approx 0.024$. Do **not** cite AI/hybrid rows with $P_{fa} = 1.0$ from that evaluation dump.

9.4 Numbers deliberately excluded

Chat-only fusion lock percentages not saved to logs (e.g. unverified $81.8\% \rightarrow 100\%$ / $79.5\% \rightarrow 97.7\%$) must not appear in submissions until re-run and archived.

10 Standards Alignment and References

1. SDA Optical Communications Terminal (OCT) Standard v3.0.1 — coarse/fine PAT stages.
2. ESA ESTOL air interface v2.2 (REQ-PHY-030) — $T_{\text{acq}} < 60$ s warm start.
3. ESA IZN-1 — ~ 2.5 mrad acquisition FOV class; μ rad-class fine tracking class.
4. Andrews & Phillips, *Laser Beam Propagation through Random Media* — C_n^2 , r_0 , scintillation, wander.
5. Kyselák et al., *Photonics* 11(6):540 (2024) — FSO acquisition scan patterns (doi:10.3390/photonics11060540). Note: literature emphasizes spirals; BeamLock default is raster.
6. Kohantorabi & Amirabadi, *Applied Optics* — deep vision YOLO FSO-PAT (doi:10.1364/ao.595667).
7. Shivkant, Jain, Ramakrishnan, arXiv:2507.20908 (IISc/QOSMIC) — probabilistic optical link-budget framing.
8. ISRO Opto-Quantum Communication (public TEC) — architecture inspiration only.

11 Limitations and Honest Claims

- Statistical channel, not phase-screen propagation.
- Link Readiness $\neq$ BER.
- Fine 4-QD/FSM is a readiness-gated mock with mild noise/latency.
- Synthetic YOLO training distribution; on-sky transfer unproven here.
- Mean pixel error is a weak detector discriminator under slew limits.
- Netlify hosts a static judge site + browser teaser; full stack is desktop `python main.py`.
- Some older docs still say “spiral”; code default is raster.

12 20-Minute Briefing Script (Detailed)

12.1 Timing map

Clock	Block	Focus
0:00–1:30	Hook	SIH26169, why FSOC needs PAT
1:30–5:00	Solution	Pipeline, UVP, architecture
5:00–9:00	Technical + physics	Channel, SNR, control law
9:00–11:30	Feasibility	Laptop ops, risks/mitigations
11:30–14:00	Impact	Who benefits, comparison table
14:00–16:00	Results	Matrix, Monte Carlo, Pd/Pfa
16:00–18:30	Demo	Video or live Calm/UAV/Stress/L/T
18:30–20:00	Close	References, honest limit, Q invite

12.2 Spoken script (expanded)

0:00–1:30. “We are Team Cadets. BeamLock addresses SIH26169: AI-based virtual camera tracking for coarse alignment of mobile FSOC terminals. Free-space optical links offer enormous bandwidth, but the beam is narrow. On a moving platform, turbulence, vibration, and noise break lock. Hardware PAT benches make algorithm iteration slow and expensive. BeamLock is a laptop simulator of the coarse PAT loop with optional Fine handoff, Classical versus AI+fusion, and reproducible metrics.”

1:30–5:00. “Solution pipeline: scene and beacon; Cn^2 physics channel; AZ/EL camera with true IFOV 6.25 microradians per pixel; Classical or AI+fusion detection; Kalman LOCKED/COASTING/LOST Link Readiness; then control—raster re-acq if lost, Fine 4-QD if handed off, else coarse slew. Uniqueness: closed-loop AZ/EL sandbox, AI inside the live loop, readiness-gated Fine, ESTOL-aligned Monte Carlo. Mapped to the PS: virtual camera, coarse alignment under mobile disturbances, AI assistance, and evidence logs.”

5:00–9:00. “Physics: UI strength maps to Cn^2 , then Fried r_0 , Rytov variance, AR beam wander, and scintillation. Link budget forms P_{rx} with geometric collection and Gaussian pointing loss, yielding SNR and fade margin. Readiness blends SNR, pointing loss, scintillation, and lock state; LOST forces zero. Control law as above. Scenarios Calm, UAV, Stress. This is Andrews-style statistics for real-time use, not a phase screen.”

9:00–11:30. “Runs on a laptop; Classical CPU fallback; CUDA optional. Ops: main.py, F1/F2/F3, T, L, B, Streamlit report. Risks: AI noise fragility mitigated by fusion; search corner-trap mitigated by raster; mean-pixel vanity metric mitigated by lock/re-acq/Tacq reporting; flight-hardware misunderstanding mitigated by honest sandbox framing.”

11:30–14:00. “Impact: faster PAT algorithm R&D, education in $\mu\text{rad}/Cn^2/r_0/\text{Tacq}$ vocabulary, reproducible evidence culture. Comparison versus typical early sims: AZ/EL plant, fusion A/B, physics channel, SNR readiness, Fine gate, raster, multi-beacon, Monte Carlo.”

14:00–16:00. “Results: UAV lock 97.7% Classical vs 100% AI; Stress re-acq 2 vs 1; mean error ~ 35 px slew-limited. Monte Carlo 40/40 within ESTOL 60 s; median Tacq 0.82 s UAV, 0.97 s Stress. Classical Stress $P_d \approx 0.67$, $P_{fa} \approx 0.024$.”

16:00–18:30. Play `beamlock_sih_demo.mp4` or live demo. Narrate overview vs FOV, cyan detection, magenta Kalman, readiness, forced blind, raster recover, AI+fusion segment.

18:30–20:00. Cite SDA OCT, ESTOL, IZN-1, Andrews, Photonics 540, Applied Optics YOLO-PAT, arXiv:2507.20908. Close: sandbox, not BER/flight plant. Invite questions.

12.3 Hard Q&A crib

- **Flight-ready?** No—virtual PAT sandbox.
- **BER?** No—readiness is handoff support.
- **Why 35 px?** Slew lag; use lock/re-acq/Tacq.
- **Spiral?** Literature yes; our default is raster.
- **Netlify full app?** No—desktop prototype; site is portal + teaser + video.

13 Reproducibility Commands

```
cd fsoc-virtual-tracker
python -m venv .venv
.venv\Scripts\activate          # Windows
pip install -r requirements.txt
python main.py
streamlit run report.py
python scripts/run_comparison.py
python scripts/monte_carlo_acquisition.py --trials 40 --scenario uav
python scripts/record_demo_video.py --fps 15 --copy-site
```

14 Conclusion

BeamLock closes a standards-aware coarse PAT loop in software: statistical atmosphere, Gaussian link SNR, Classical/AI+fusion detection, Kalman tracking, raster re-acquisition, and readiness-gated Fine mock, with reproducible Monte Carlo and A/B evidence. For SIH26169 it demonstrates an AI-assisted virtual camera tracking system for coarse alignment of mobile FSOC terminals, with explicit physics assumptions and explicit non-claims.

End of document.